



Arithmetic Series

How quickly can you add the numbers from 1 to 100 in your head?

It is said that in elementary school, the famous mathematician Carl Friedrich Gauss added the integers from 1 to 100 in seconds by recognizing that the sum was 50 pairs of numbers, each of which added to 101.

Arithmetic Series

Definition

An **arithmetic series** is a series that results from adding the terms of an arithmetic sequence.

► **Remember** A **series** is the sum of consecutive terms of a sequence.

For example, if you start with the arithmetic sequence

$$5, 8, 11, 14, \dots,$$

then you can define the corresponding series:

$$5 + 8 + 11 + 14 + \dots$$

Because the sequence goes on forever, the sum also goes on forever.

In general, if a_n represents the terms of an arithmetic sequence, then the sum of all the terms of the sequence is

$$S = a_1 + a_2 + a_3 + a_4 + \dots$$

When you are interested in the sum of the first n terms of the arithmetic sequence, you can write the n th partial sum of the series as follows:

$$S_n = a_1 + a_2 + a_3 + a_4 + \dots + a_{n-2} + a_{n-1} + a_n.$$

Finding the Sum of the First n Terms of an Arithmetic Series

A general formula for the sum, S_n , of the first n terms of an arithmetic series can be seen with the help of an example.

sums of pairs

 $20 + 23 = 43$
 $17 + 26 = 43$
 $14 + 29 = 43$
 $11 + 32 = 43$
 $8 + 35 = 43$
 $5 + 38 = 43$

You can see that each of these six pairs sums to 43. This is the type of pattern that Gauss recognized and used to calculate the sum so quickly.

Arithmetic Series Formula

The sum of the first n terms of an arithmetic series with first term a_1 and n th term a_n is given by

$$S_n = \frac{n}{2} (a_1 + a_n).$$

EXAMPLE 1

A Find the sum of the first 100 positive even integers.

SOLUTION

There are $n = 100$ terms. Also, $a_1 = 2$ and $a_n = 200$.

$$\begin{aligned} S_{100} &= \frac{100}{2} (2 + 200) \\ &= 50 \cdot 202 = 10,100 \end{aligned}$$

So the sum of the first 100 positive even integers is 10,100.

B Find the sum: $\sum_{i=1}^{17} 2i + 5$.

SOLUTION

There are $n = 17$ terms. Find a_1 and a_n .

$$a_1 = 2 \cdot 1 + 5 = 7$$

$$a_{17} = 2 \cdot 17 + 5 = 39$$

Now find S_{17} .

$$\begin{aligned} S_{17} &= \frac{17}{2} (7 + 39) \\ &= 8.5 \cdot 46 = 391 \end{aligned}$$

So $\sum_{i=1}^{17} (2i + 5) = 391$. ■

► **Think About It** To find the sum of the first n terms of an arithmetic series by using the formula

$$S_n = \frac{n}{2} (a_1 + a_n),$$

you need to know the number of terms, n , the first term, a_1 , and the last term, a_n .

An Alternate Formula for the Sum of the First n Terms of an Arithmetic Series

If you substitute $a_1 + (n - 1)d$ for a_n in the formula $S_n = \frac{n}{2} (a_1 + a_n)$, you get an alternate formula for the sum of the first n terms of an arithmetic series.

$$S_n = \frac{n}{2} (a_1 + a_n) = \frac{n}{2} (a_1 + a_1 + (n - 1)d) = \frac{n}{2} (2a_1 + (n - 1)d)$$

Alternate Arithmetic Series Formula

The sum of the first n terms of an arithmetic series with first term a_1 and common difference d is given by

$$S_n = \frac{n}{2} (2a_1 + (n-1)d).$$

EXAMPLE 2

A Find the sum of the first 20 terms: $-35 + (-30) + (-25) + (-20) + \dots$

SOLUTION

The first term is $b_1 = -35$, and the constant difference is $d = 5$. So the sum of the first 20 terms can be found as shown:

$$\begin{aligned} S_n &= \frac{n}{2} (2b_1 + (n-1)d) \\ &= \frac{20}{2} [2(-35) + (20-1)5] \\ &= 10(-70 + 19 \cdot 5) \\ &= 10 \cdot 25 \\ &= 250 \end{aligned}$$

So the sum of the first 20 terms is 250.

► **Remember** The sum of the first n terms of a series is called the n th partial sum. In Example 2A, the 20th partial sum is 250.

B Find the sum of the first 14 terms of the arithmetic series where $c_1 = 32,000$ and common difference $d = 1500$.

SOLUTION

There are $n = 14$ terms in the series. The first term is $c_1 = 32,000$, and the constant difference is $d = 1500$. So the sum of the first 14 terms is

$$\begin{aligned} S_n &= \frac{n}{2} (2c_1 + (n-1)d) \\ &= \frac{14}{2} [2 \cdot 32,000 + (14-1)1500] \\ &= 7(64,000 + 19,500) \\ &= 7 \cdot 83,500 \\ &= 584,500 \end{aligned}$$

So the sum of the first 14 terms is 584,500. ■

Graphing the Partial Sums of an Arithmetic Series

EXAMPLE 3

Graph the first five partial sums of the arithmetic series.

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + \dots$$

SOLUTION

Make a table of the five partial sums and then plot the points.

n	S_n
1	1
2	$1 + 2 = 3$
3	$1 + 2 + 3 = 6$
4	$1 + 2 + 3 + 4 = 10$
5	$1 + 2 + 3 + 4 + 5 = 15$

